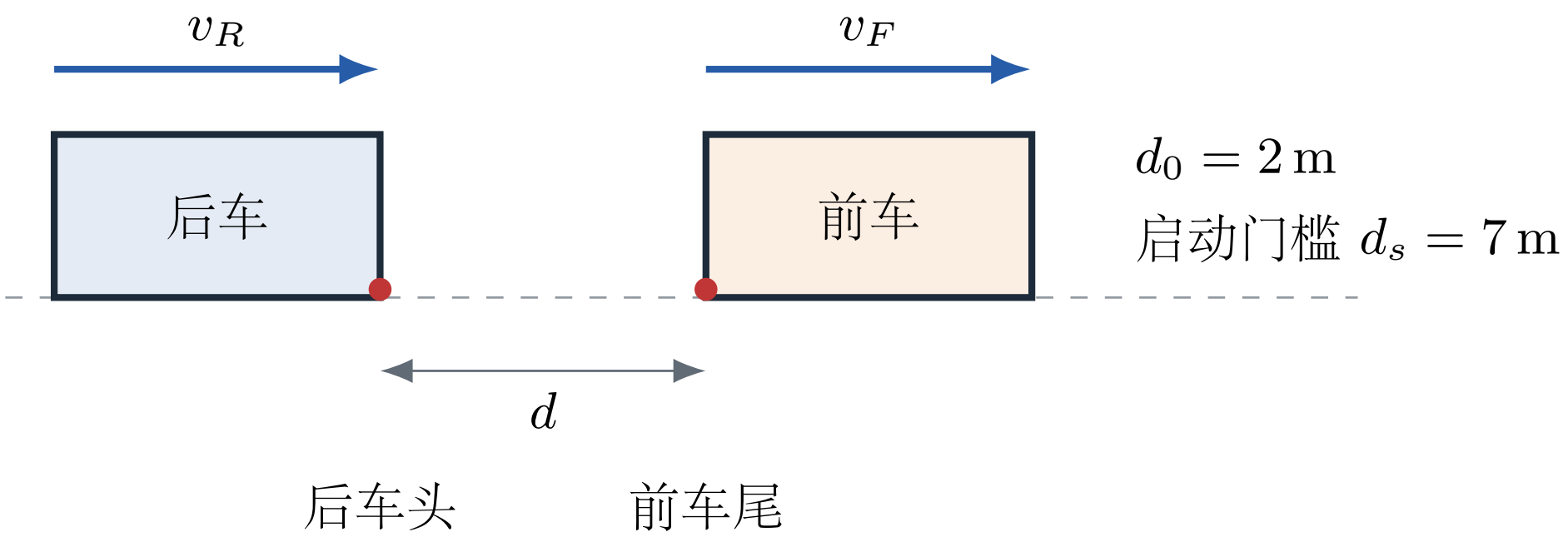
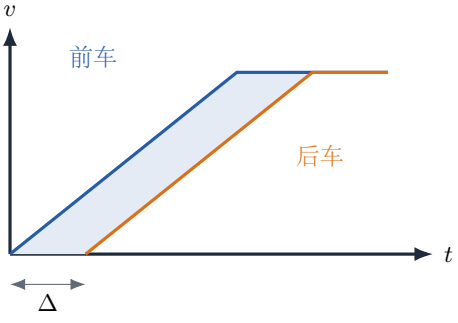
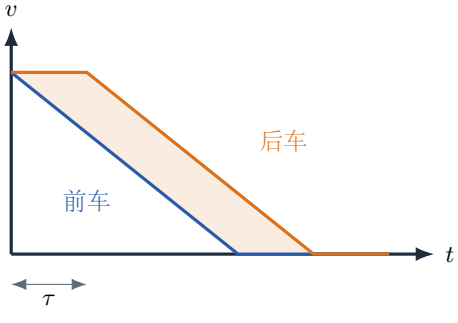

v_R
v_F
后车
前车
$d_0 = 2\,\mathrm{m}$
启动门槛 $d_s = 7\,\mathrm{m}$
d
后车头
前车尾

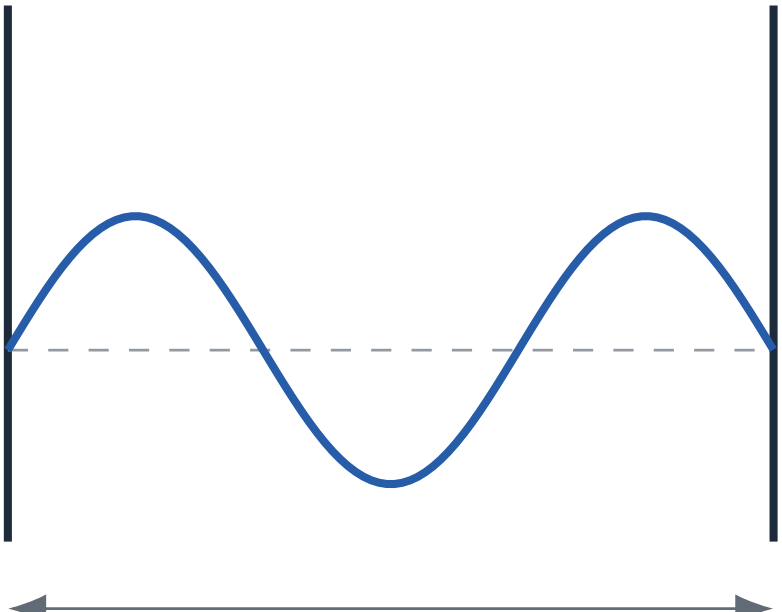


启动：$\Delta d = v\Delta$

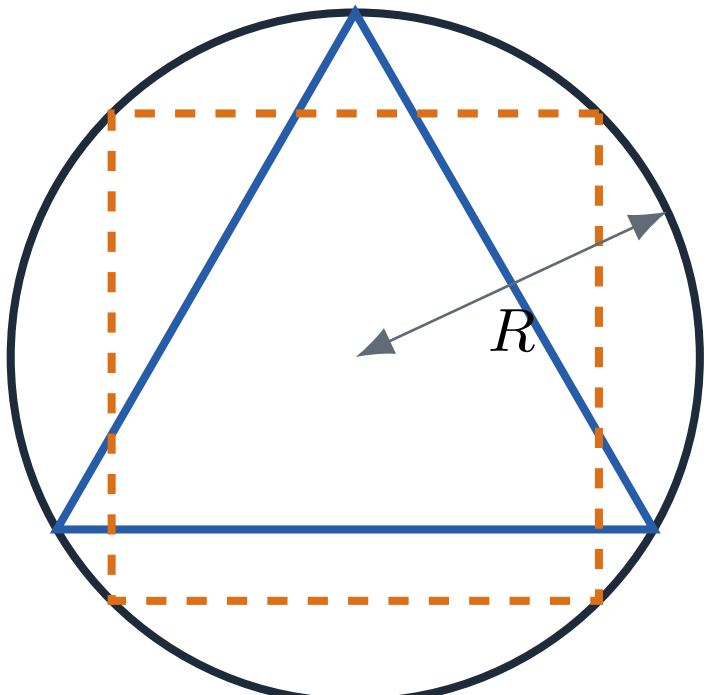


制动：$\Delta d = -v\tau$

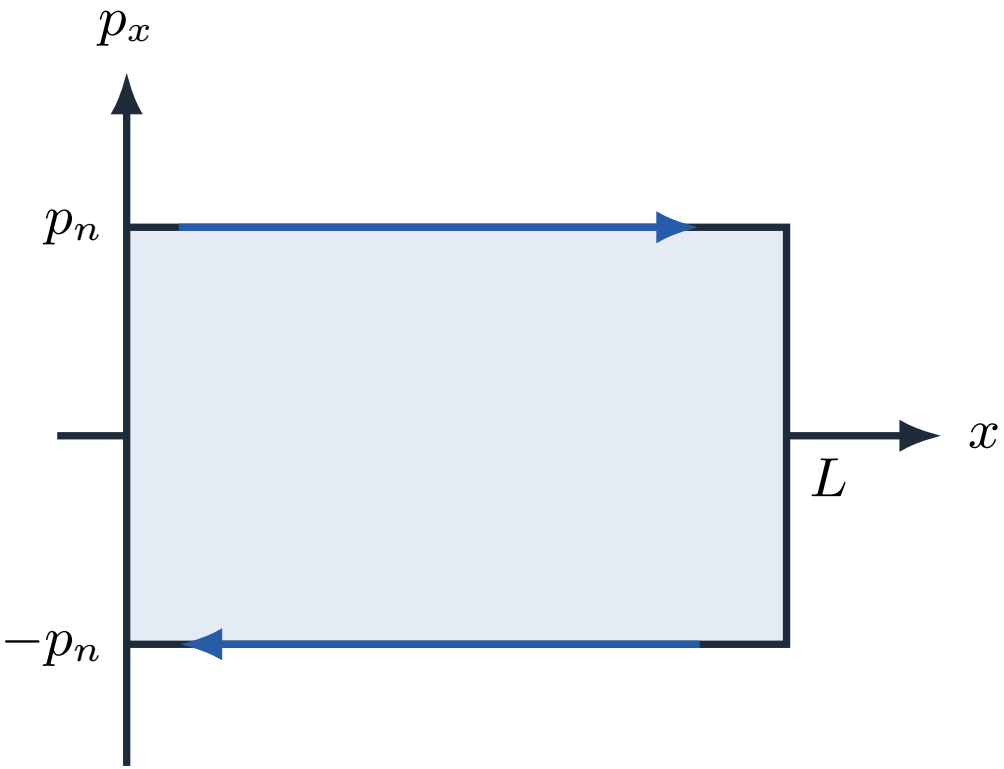
$n = 3$ 驻波示例



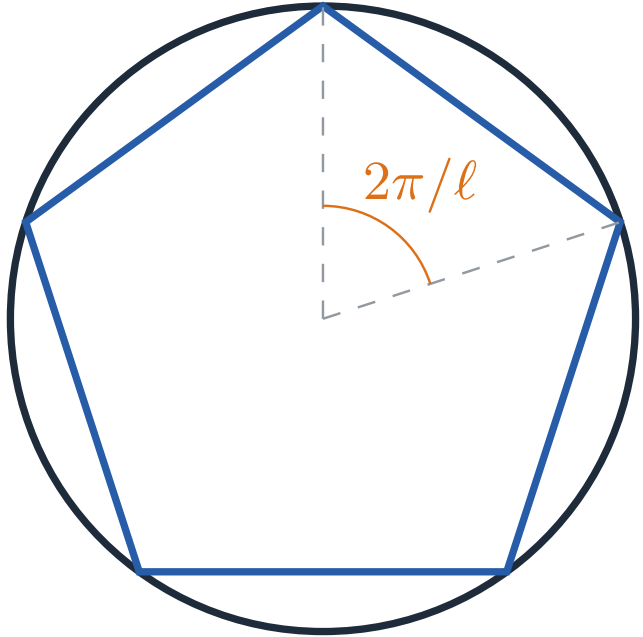
$L = n\lambda_n/2$



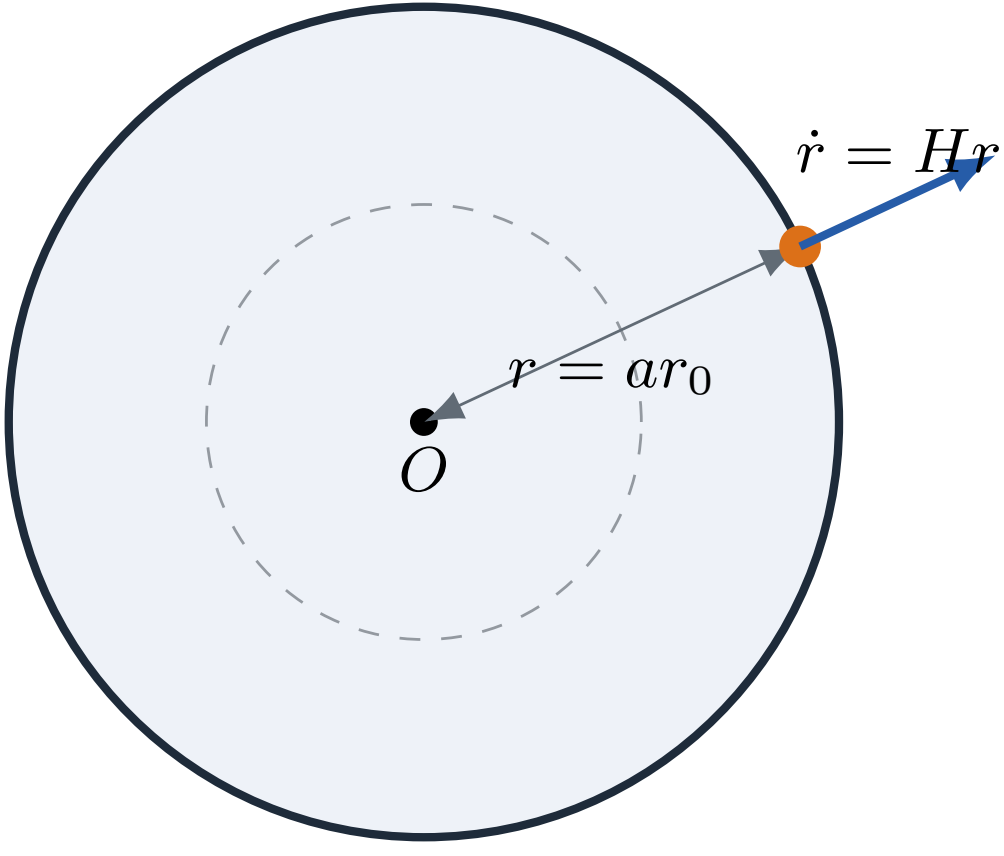
闭合轨道的传播相位条件



$$\mathcal{A} = 2Lp_n = nh$$



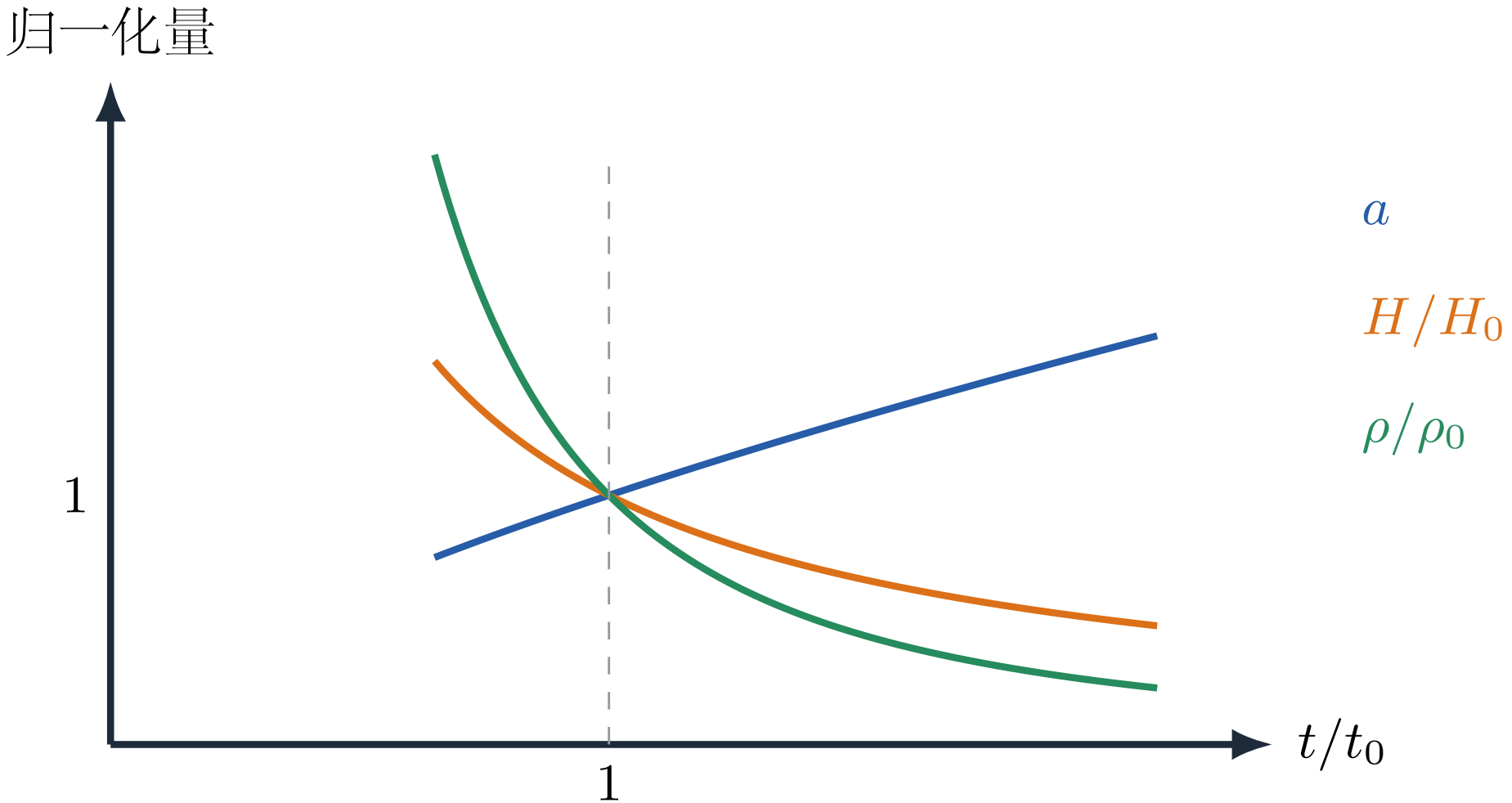
$$P_\ell = 2\ell R \sin(\pi/\ell)$$



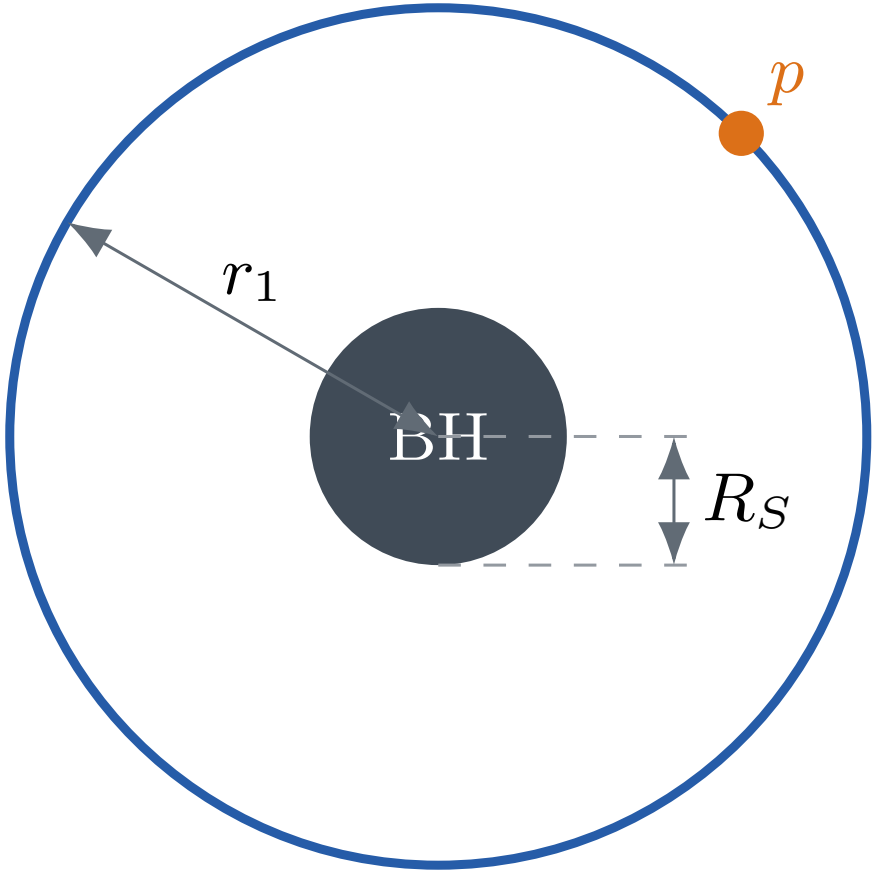
$$M = \frac{4\pi}{3}\rho r^3 = \text{const}$$

$$\rho = \rho_0 a^{-3}$$

共动球跟踪同一批物质



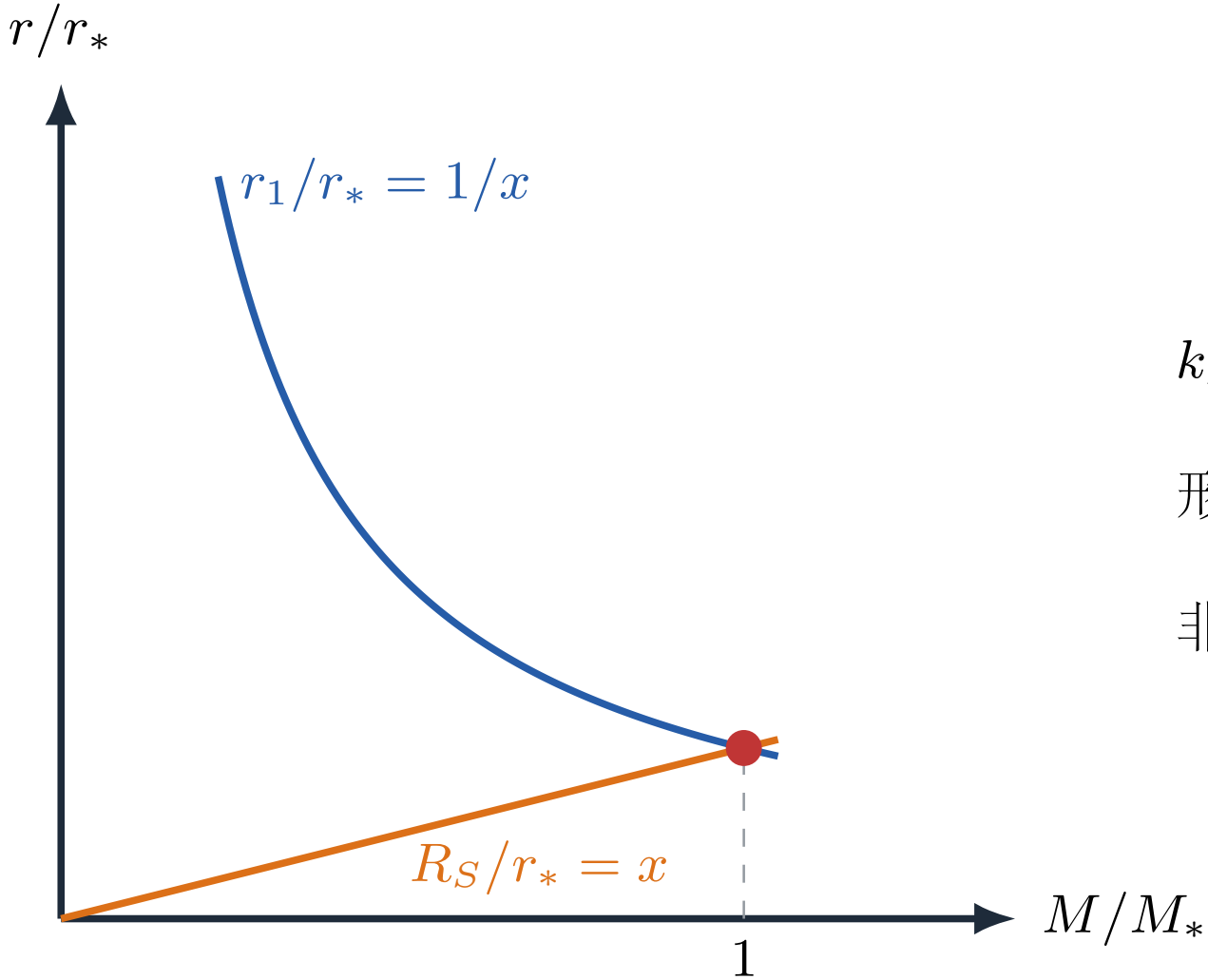
$$a = (t/t_0)^{2/3}, \quad H \propto t^{-1}, \quad \rho \propto t^{-2}$$



$$r_1 = \hbar^2/(GMm_p^2)$$

$$R_S = 2GM/c^2$$

图为牛顿—波尔估算模型

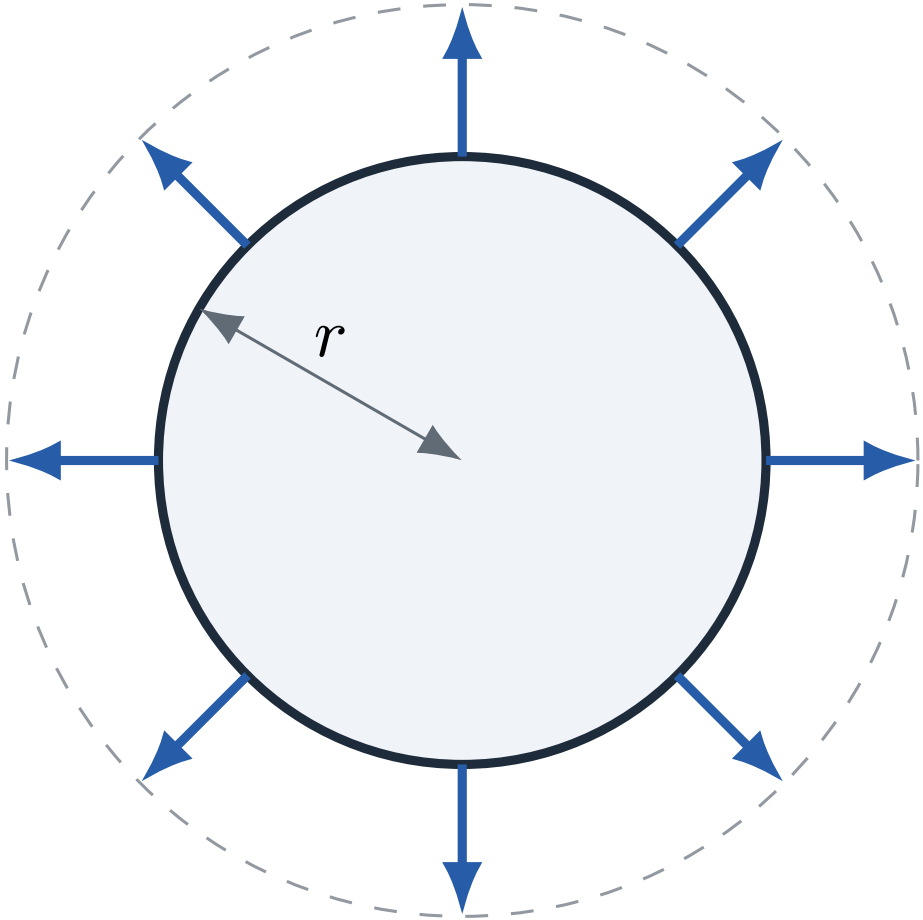


$k_B T/E_{\rm ion} \simeq 0.183/x^3$

形式交点：$v_1 = c/\sqrt{2}$

非相对论近似在交点失效

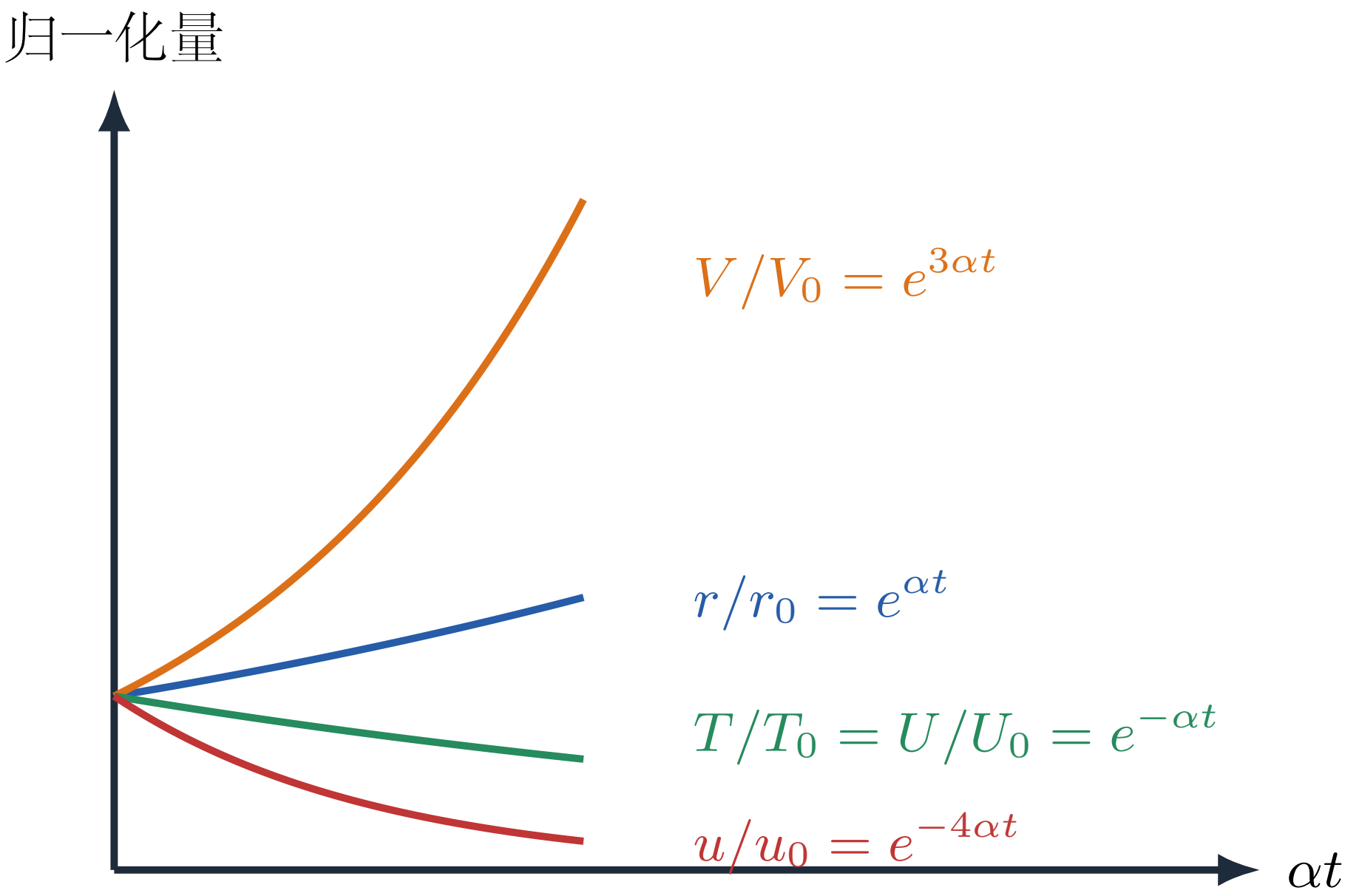
$x = M/M_*$；等号为形式边界



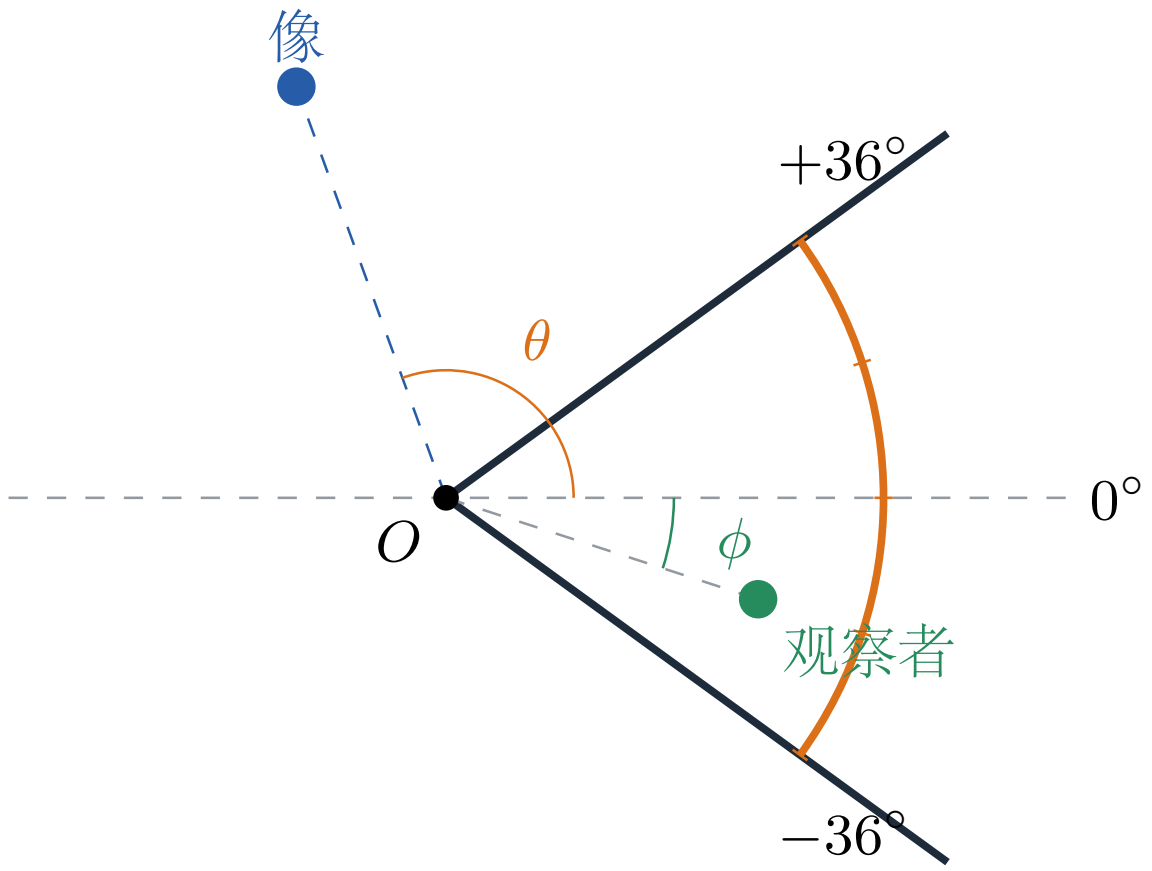
$$u = \beta T^4, \quad P = u/3$$

$$U = uV, \quad \dot{r} = \alpha r$$

绝热准静态膨胀

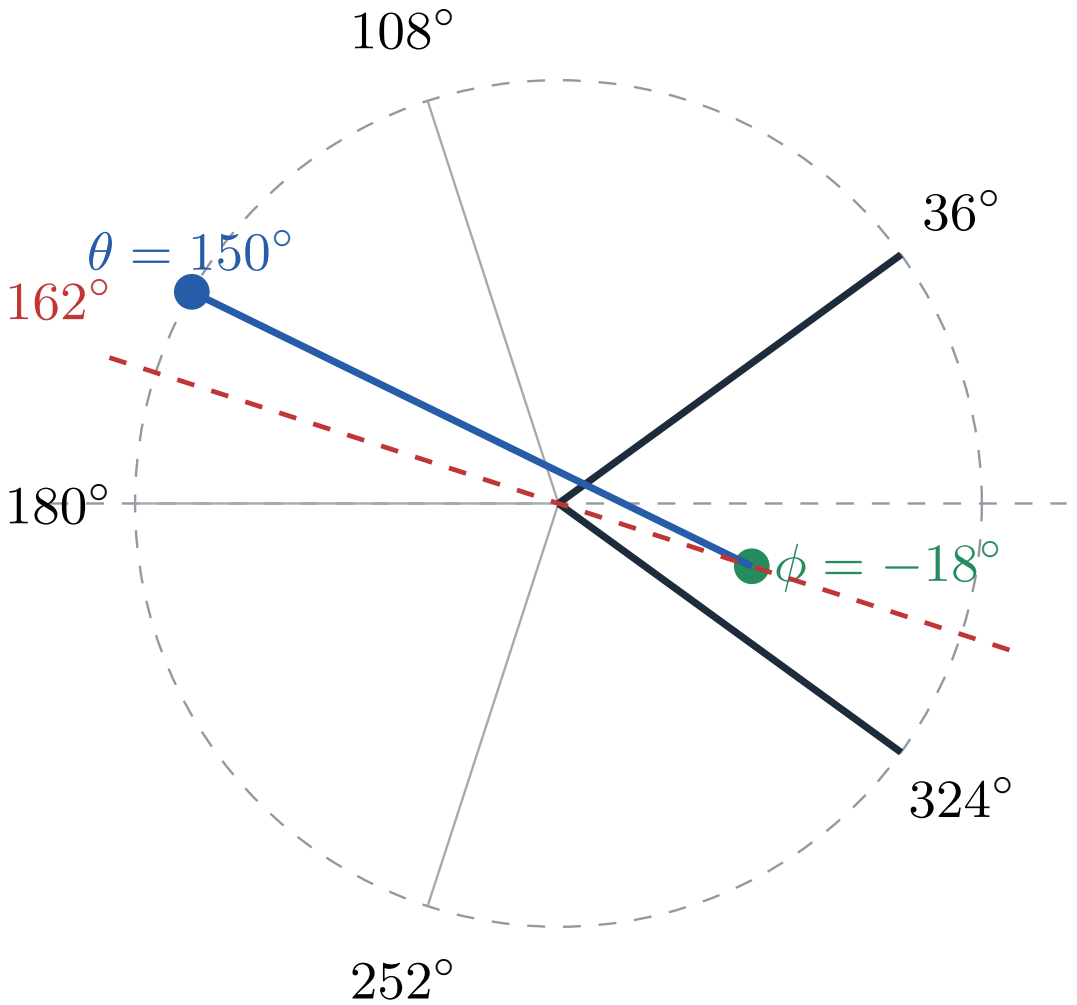


$$Tr = T_0 r_0, \quad S = \frac{4}{3}\beta V T^3 = \text{const}$$



θ, ϕ 共用角平分线基准

原刻度：$-36^\circ < \varphi < 36^\circ$

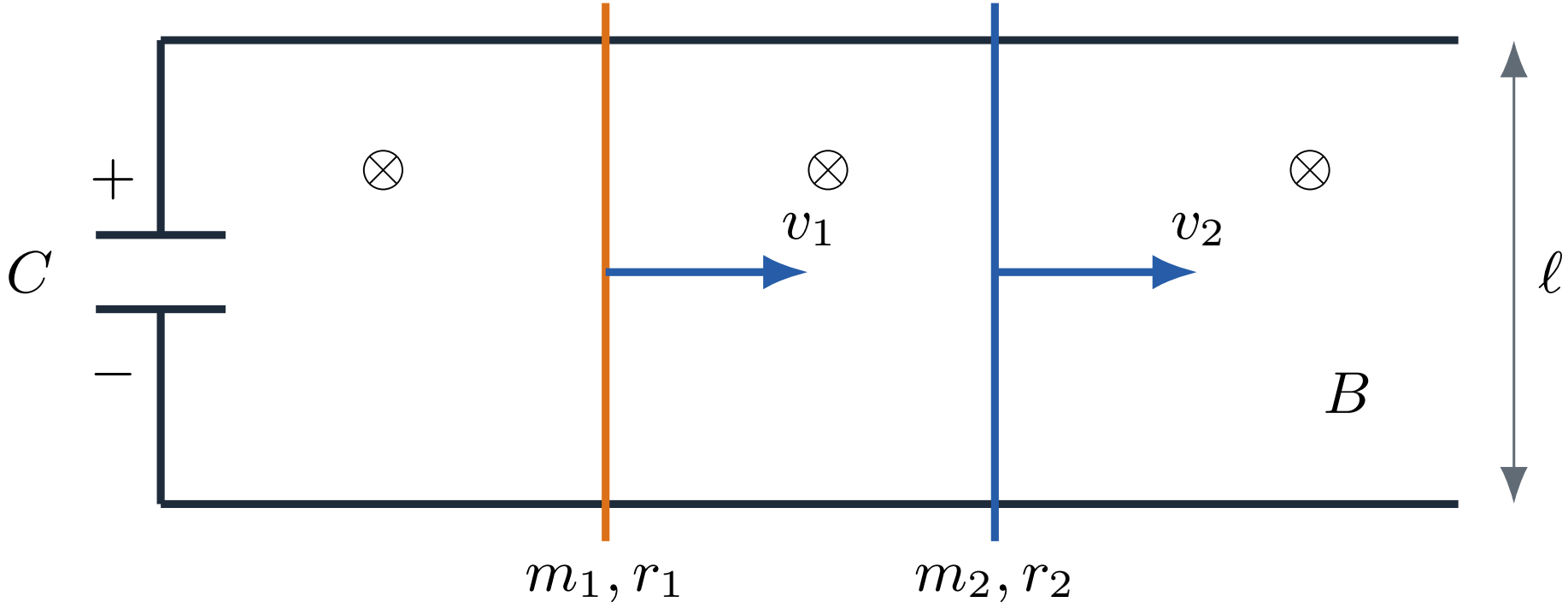


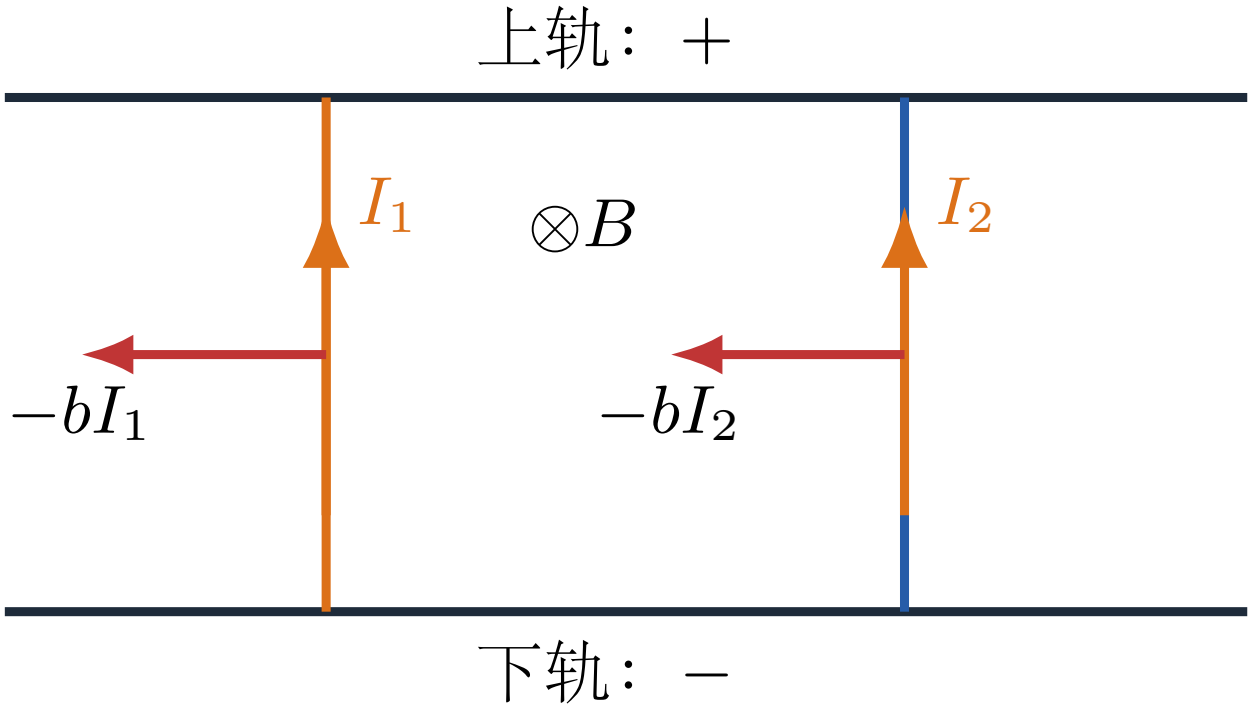
正向：$\theta < 180° + \phi$

负向：$\theta > 180° + \phi$

等号：连线穿过镜面顶点

第二区：$\varphi = \pm(\theta - 144°)$


+
C
−
v_1
v_2
B
ℓ
m_1, r_1
m_2, r_2

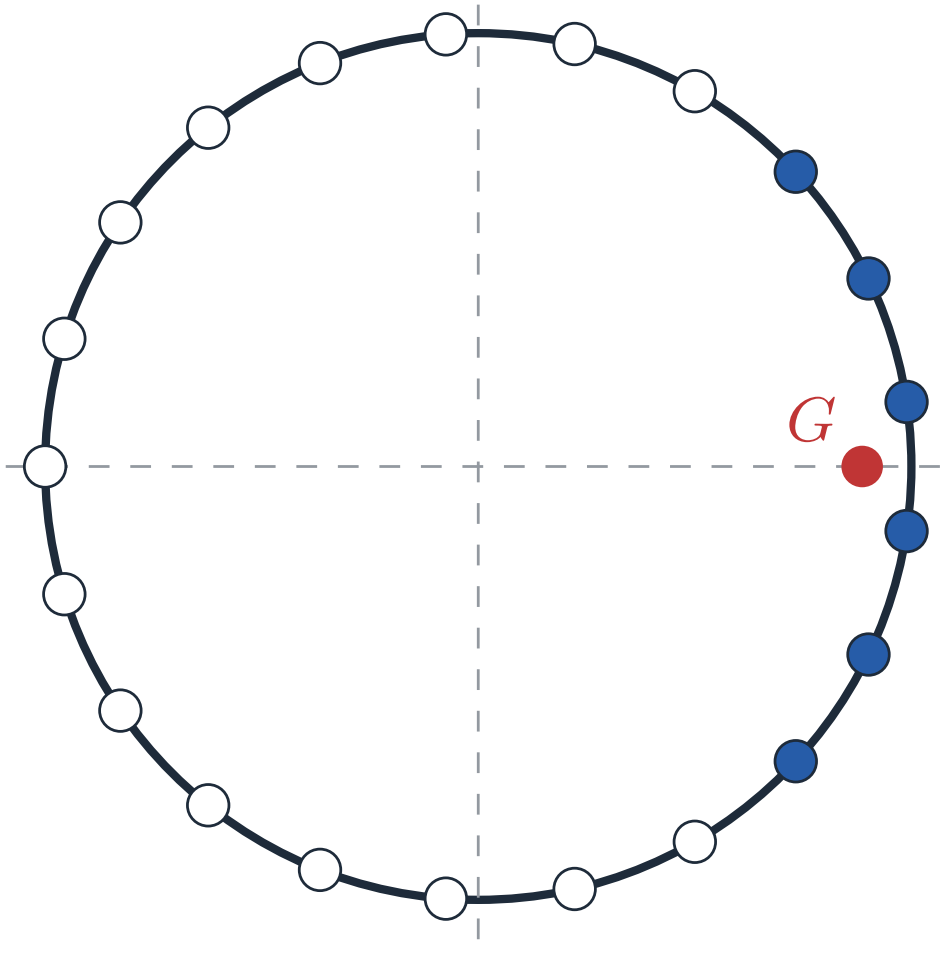


$$r_i I_i = bv_i - V$$

$$m_i \dot{v}_i = -bI_i$$

$$C\dot{V} = I_1 + I_2$$

$$m_1 v_1 + m_2 v_2 + bCV = \text{const}$$

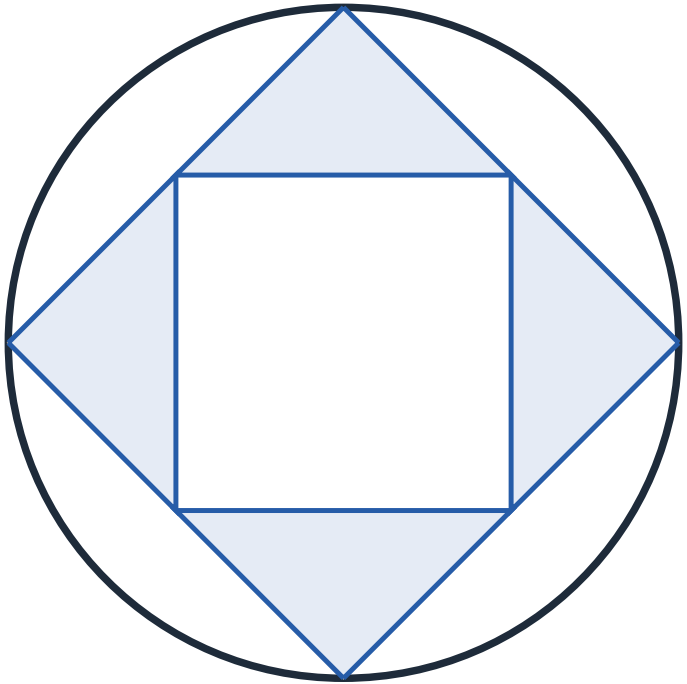


实心：6 条承压腿

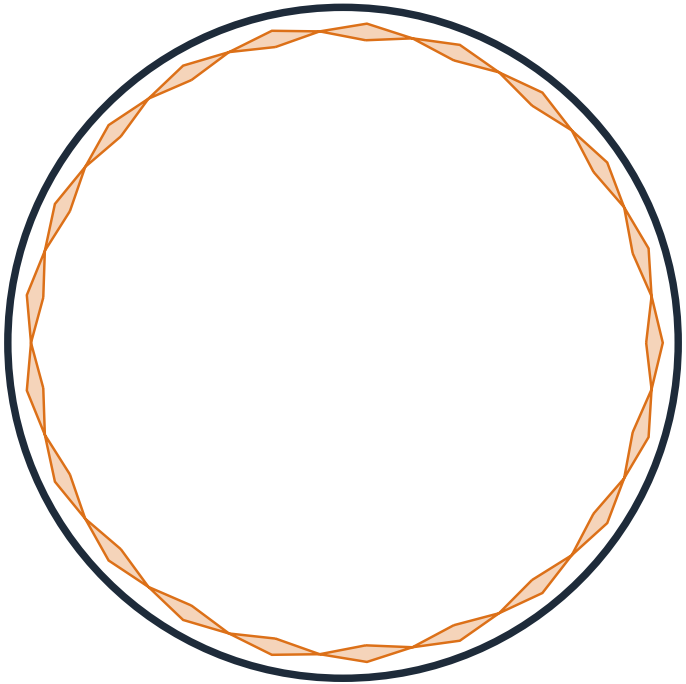
空心：15 条悬空腿

$$F_i = k \max(\delta_i, 0)$$

$$\delta_i = a + b_x \cos\vartheta_i + b_y \sin\vartheta_i$$



$N = 4 : \quad S_1 = r^2$



$N = 21, m = 15 : \quad S_2 \simeq 0.14463434 r^2$